

Week 2 / Fire Wall

Abstract

This project implements a Personal Firewall System using Python to monitor, analyze, and control network traffic in real time. The system captures packets using the Scapy library and applies rule-based filtering to allow or block traffic based on IP address, port, and protocol. It provides both CLI and GUI interfaces along with logging and statistics.

Introduction

With increasing cyber threats, protecting systems from unauthorized access is essential. Firewalls act as a security layer that filters network traffic. This project builds a software-based firewall that enables users to define rules and monitor packets in real time.

Tools Used

- Python
- Scapy Library
- Tkinter
- Threading, Datetime Modules

Steps Involved in Building the Project

1. Requirement Analysis
2. Packet Capture using Scapy
3. Rule-Based Filtering Implementation
4. Logging and Statistics
5. CLI Development
6. GUI Development
7. Testing and Debugging

Conclusion

The project demonstrates how network packets can be monitored and controlled using Python. It improves understanding of network security concepts and can be extended with advanced features like intrusion detection and AI-based threat analysis.